Game Room: Offline / LAN Multiplayer Games

Seven multiplayer games that run with **no internet**. One device hosts a tiny local server, and everyone else joins from their browser over the same Wi-Fi or phone hotspot. The game traffic itself goes peer-to-peer (WebRTC), and the host's browser acts as the match host.

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1. Components

File	What it is
`server.js`	Zero-dependency Node.js server. Serves the pages, runs each game's lobby, relays WebRTC signalling and stores the leaderboard.
`index.html`	**Game Room** launcher. Shows all seven games as cards, plus a "Last played /

File	What it is
	Play again" shortcut (saved in the browser).
`checkers.html`	**Checkers**: jump diagonally and capture every piece.
`bead16.html`	**Sixteen Beads** (Sholo Guti): move along the lines, jump to capture.
`carrom.html`	**Carrom**: flick the striker and pocket your coins.
`ludo.html`	**Ludo**: roll the dice and race your tokens home (2 to 5 players).
`snake_io.html`	**Snake.io**: eat, grow and outlast everyone in the arena.
`aamjam_lichi.html`	**Aam Jam Kanthal Lichi**: 4-match card game, Bengali interface. Collect four of a kind first.
`hundred.html`	**Hundred**: race to one hundred (card game).
`scores.json`	Created automatically by `server.js` when the first

File	What it is
	score is saved (leaderboard).
`README.md`	This file.

Keep **all files in the same folder**. The launcher links to the games by file name.

Requirements

- Node.js 16 or newer on the host device only. No `npm install` is needed.
- A modern browser on every device (Chrome, Firefox, Safari, Edge).
- All devices on the **same network**: a router, a phone hotspot, or Wi-Fi with no internet all work.

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2. How it works

What you do	Needs `server.js` ?	Who can play
Solo vs computer / bots (Checkers, Sixteen Beads, Carrom, Ludo,	No	Just you

What you do	Needs `server.js`?	Who can play
Hundred, Snake.io practice)		
Quick match, Create room, Join room	**Yes**	Any devices on the same network
Aam Jam Lichi: Create Room / Join Room, with optional bots	**Yes**	Any devices on the same network

- Every game has its own separate lobby, so rooms from one game never show up in another.
- If `server.js` is not running, the online buttons can't connect. Solo modes still work.
- Rooms are public (listed) or private (join by room code or invite link).

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3. Setup: Android phone as host (Termux)

One-time (needs internet once to install Node):

```
pkg update && pkg upgrade
pkg install nodejs
termux-setup-storage
mkdir -p ~/games && cd ~/games
cp ~/storage/downloads/*.html .
cp ~/storage/downloads/server.js .
```

Allow storage access when Android asks. If the files are not in Downloads, change the path in the `cp` commands.

Every time you play:

```
cd ~/games
termux-wake-lock
node server.js
```

The server prints something like:

```
Local games server running (no internet
needed)

This device : http://localhost:3000
Other devices: http://192.168.43.1:3000
(wlan0)
```

If it cannot detect your IP it says so. Then run `ifconfig` or `ip addr` yourself. When the host phone is the hotspot, friends can usually use `http://192.168.43.1:3000`.

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4. Setup: PC / laptop as host (Windows, Mac, Linux)

Install Node.js, put all files in one folder, then:

```
cd path/to/folder
node server.js
```

Change the port if 3000 is busy:

```
PORT=8080 node server.js          # Mac / Linux / Termux
set PORT=8080 && node server.js   # Windows (cmd)
```

Allow Node.js through the firewall if asked.

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5. Playing with no internet

1. **Host phone:** turn on the **hotspot**. Mobile data can stay off.
2. **Other players:** join that hotspot. Ignore the "no internet" warning and stay connected.
3. **Host:** run `node server.js`.
4. **Host:** open the printed **Other devices** address in the browser. Do not use `localhost`, so the host

connects the same way everyone else does.

5. **Host:** pick a game on the Game Room page, then tap **Create room** (or **Quick match**). Share the room code.
6. **Others:** open the same address, pick the same game, then tap **Join room** and choose the room or enter the code. Invite links (`.../ludo.html#room=ABCD`) also work.
7. **Host:** press **Start** once everyone has joined.

Aam Jam Lichi: tap **Create Room**, share the code, and use **+ bot / − bot** to fill empty seats. Others tap **Join Room** and enter the code.

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6. Command cheat sheet

Task	Command
Install Node (Termux)	`pkg install nodejs`
Grant storage (Termux)	`termux-setup-storage`
Start server	`node server.js`
Start on another port	`PORT=8080 node server.js`

Task	Command
Keep Termux awake	`termux-wake-lock`
Release wake lock	`termux-wake-unlock`
Show your IP (Termux / Linux)	`ip -4 addr` or `ifconfig`
Show your IP (Mac)	`ifconfig`
Show your IP (Windows)	`ipconfig`
Stop server	`Ctrl + C`
Reset leaderboard	stop the server, delete `scores.json`

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7. Troubleshooting

Problem	Fix
Page does not load on other devices	Check the server is still running, the IP is typed exactly, and everyone is on the same hotspot/Wi-Fi.

Problem	Fix
Players can't see each other's rooms	Turn off **AP / client isolation** in the hotspot settings.
Phone keeps dropping the Wi-Fi ("no internet")	Turn off mobile data on that phone, or choose "stay connected" when Android asks.
Log says "could not connect"	The two devices can't reach each other directly. Check isolation, and make sure the host opened the network address, not `localhost`.
Two tabs on one phone fail to connect	Phone browsers freeze hidden tabs. Use two devices, two different browsers in split-screen, or a PC with two windows.
Online buttons fail, solo works	`server.js` is not running or the address is wrong.
Voice chat or microphone is blocked	Browsers only allow the microphone on HTTPS or `localhost`. Plain `http://192.168.x.x` is blocked. In Chrome you can open `chrome://flags/#unsafely-`

Problem	Fix
	`treat-insecure-origin-as-secure`, add the server address, and relaunch. Text chat is not affected.
Aam Jam Lichi text looks different	The page asks Google Fonts for Bengali fonts. With no internet it falls back to your phone's built-in Bengali font. It still works.
Server stops when Termux is closed	Run `termux-wake-lock`, keep Termux open, and exclude Termux from battery optimisation.
Port already in use	Use another port: `PORT=8080 node server.js`.

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8. Notes

- **The host's browser runs the match.** The host must keep the game tab open and in front while the match runs.
- **Privacy.** Nothing leaves your network. There are no accounts or API keys, and the games make no external requests (except the optional fonts above).

- **Why the server rewrites IPs.** Browsers hide LAN addresses behind random `xxxx.local` names that many hotspots can't resolve. `server.js` replaces them with each player's real IP so connections succeed.
- **Security.** `server.js` and `scores.json` are never served to browsers. Run it only on networks you trust, since anyone on the network can open the pages.